

Understanding Op Amp Parameters

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13.1 Introduction

This chapter is about op amp data sheet parameters. There are usually three main sections of electrical tables in op amp data sheets.

Absolute maximum ratings are those limits beyond which the life of individual devices may be impaired and are never to be exceeded in service or testing. Limits, by definition, are maximum ratings, so if double ended limits are specified, the term is defined as a range (e.g., operating temperature range).

Recommended operating conditions have a similarity to maximum ratings, in that operation outside the stated limits could cause unsatisfactory performance.

Recommended operating conditions, however, do not carry the implication of device damage if they are exceeded.

Electrical characteristics are measurable electrical properties of a device inherent in its design. They are used to predict the performance of the device as an element of an electrical circuit. The measurements that appear in the electrical characteristics tables are based on the device being operated within the recommended operating conditions.

[Table 13.1](#) lists op amp condition and parameter abbreviations plus their corresponding description and units. [Select active links](#) to obtain additional information. As the name of the table implies, some of it is composed of parameters and some test conditions. Test conditions are conditions placed on the op amp when the parameters are measured. Some abbreviations are used for both a condition and a parameter. Units listed in the

Table 13.1: Op Amp Conditions and Parameters

Abbreviation	Parameter	Units	Abbreviation	Parameter	Units
$\alpha_{I_{IO}}$	Temperature coefficient of input offset current	A/°C	SR	Slew rate	V/S
$\alpha_{V_{IO}}$ or $\alpha_{V_{IO}}$	Temperature coefficient of input offset voltage	V/°C	T_A	Operating temperature	°C
A_D	Differential gain error	%	t_{DIS} or $t_{(off)}$	Turn off time (shutdown)	s
A_m	Gain margin	dB	t_{EN} or $t_{(on)}$	Turn on time (shutdown)	s
A_{OL}	Open loop voltage gain	dB	t_f	Fall time	s
A_V	Large signal voltage amplification (gain)	dB	THD	Total harmonic distortion	%
A_{VD}	Differential large signal voltage amplification	dB	THD + N	Total harmonic distortion plus noise	%
B_1	Unity gain bandwidth	Hz	T_J	Maximum junction temperature	°C
B_{OM}	Maximum output swing bandwidth	Hz	t_r	Rise time	s
BW	Bandwidth	Hz	t_s	Settling time	s
C_I	Input capacitance	F	T_S or T_{stg}	Storage temperature	°C
C_{ic} or $C_{i(c)}$	Common mode input capacitance	F	V_{CC}, V_{DD}	Supply voltage	V
C_{id}	Differential input capacitance	F	V_I	Input voltage range	V
C_L	Load capacitance	F	V_{IC}	Common mode input voltage	V
$\Delta V_{DD\pm}$ (or $CC\pm$)/ ΔV_{IO} , or k_{SVS}	Supply voltage sensitivity	dB	V_{ICR}	Input common mode voltage range	V

Table 13.1: Op Amp Conditions and Parameters (Cont'd)

Abbreviation	Parameter	Units	Abbreviation	Parameter	Units
CMRR or k_{CMR}	Common mode rejection ratio	dB	V_{ID}	Differential input voltage	V
f	Frequency	Hz	V_{DIR}	Differential Input voltage range	V
GBW	Gain bandwidth product	Hz	$V_{\text{IH-SHDN}}$ or $V_{\text{(ON)}}$	Turn on voltage (shutdown)	V
$I_{\text{CC-SHDN}}$, $I_{\text{DD-SHDN}}$	Supply current (shutdown)	A	$V_{\text{IL-SHDN}}$ or $V_{\text{(OFF)}}$	Turn off voltage (shutdown)	V
I_{CC} , I_{DD}	Supply current	A	V_{IN}	Input voltage (DC)	V
I_{I}	Input current range	A	V_{IO} , V_{OS}	Input offset voltage	V
I_{IB}	Input bias current	A	V_n	Equivalent input noise voltage	$\text{V}/\sqrt{\text{Hz}}$
I_{IO}	Input offset current	A	$V_{\text{N(PP)}}$	Broadband noise	V P-P
I_n	Input noise current	$\text{A}/\sqrt{\text{Hz}}$	V_{OH}	High level output voltage	V
I_{O}	Output current	A	V_{OL}	Low level output voltage	V
I_{OL}	Low level output current	A	$V_{\text{OM}\pm}$	Maximum peak to peak output voltage swing	V
I_{OS} or I_{SC}	Short circuit output current	A	$V_{\text{O(PP)}}$	Peak to peak output voltage swing	V
CMRR or k_{CMR}	Common mode rejection ratio	dB	$V_{\text{(STEP)PP}}$	Step voltage peak to peak	V
k_{SVR}	Supply rejection ratio	dB	X_T	Crosstalk	dB
k_{SVS}	Supply voltage sensitivity	dB	Z_o	Output impedance	Ω
P_D	Power dissipation	W	Z_t	Open loop transimpedance	Ω

(Continued)

Table 13.1: Op Amp Conditions and Parameters (Cont'd)

Abbreviation	Parameter	Units	Abbreviation	Parameter	Units
PSRR	Power supply rejection ratio	dB	Φ_D	Differential phase error	°
θ_{JA}	Junction to ambient thermal resistance	°C/W	Φ_m	Phase margin	°
θ_{JC}	Junction to case thermal resistance	°C/W		Bandwidth for 0.1 dB flatness	Hz
r_i	Input resistance	Ω		Case temperature for 60 s	°C
$r_{id}, r_{i(d)}$	Differential input resistance	Ω		Continuous total dissipation	W
R_L	Load resistance	Ω		Differential gain error	%
R_{null}	Null resistance	Ω		Differential phase error	°
r_o	Output resistance	Ω		Duration of short circuit current	s
R_S	Signal source resistance	Ω		Input offset voltage long term drift	V month
				Lead temperature for 10 or 60 s	°C

units column of Table 13.1 are part of the standard SI units of measure. Multiplier prefixes such as p (pico) and M (mega) are often used in data sheets.

13.2 Temperature Coefficient of the Input Offset Current, $\alpha_{I_{IO}}$

The temperature coefficient of the input offset current, $\alpha_{I_{IO}}$, is defined as the ratio of the change in input offset current to the change in the die temperature. This is an average value for the specified temperature range.

The term αI_{IO} specifies the expected input offset current drift over temperature. Its units are microamperes per degree Centigrade. I_{IO} is measured at the temperature extremes of the part, and αI_{IO} is computed as $\Delta I_{IO}/\Delta^\circ\text{C}$.

Normal aging in semiconductors causes changes in the characteristics of devices. The input offset voltage long term drift specifies how I_{IO} is expected to change with time. Its units are amps per month.

13.3 Temperature Coefficient of Input Offset Voltage, αV_{IO} or α_{VIO}

The temperature coefficient of input offset voltage, αV_{IO} or α_{VIO} , is defined as the ratio of the change in input offset voltage to the change in the die temperature. This is an average value for the specified temperature range.

The term αV_{IO} specifies the expected input offset drift over temperature. Its units are volts per degree Centigrade. V_{IO} is measured at the temperature extremes of the part, and αV_{IO} is computed as $\Delta V_{IO}/\Delta^\circ\text{C}$.

Normal aging in semiconductors causes changes in the characteristics of devices. The input offset voltage long term drift specifies how V_{IO} is expected to change with time. Its units are microvolts per month.

13.4 Differential Gain Error, A_D

The differential gain error parameter, A_D , is defined as the change in AC gain with a change in DC level. The AC signal is 40 IRE (0.28 V PK), and the DC level change is ± 100 IRE (± 0.7 V). Typically tested at 3.58 MHz (NTSC) or 4.43 MHz (PAL) carrier frequencies. It is represented in units of percent.

13.5 Gain Margin Parameter, A_m

Gain margin, A_m , is defined as the absolute value of the difference in gain between the unity gain point and the gain at the -180° phase shift point. It is measured for an open loop and expressed in units of decibels.

Gain margin (A_m) and phase margin (Φ_m) are different ways of specifying the stability of the circuit. Since rail to rail output op amps have higher output impedance, a significant phase shift is seen when driving capacitive loads. This extra phase shift

erodes the phase margin; and for this reason, most CMOS op amps with rail to rail outputs have limited ability to drive capacitive loads.

Figure 13.1 shows the gain margin graphically.

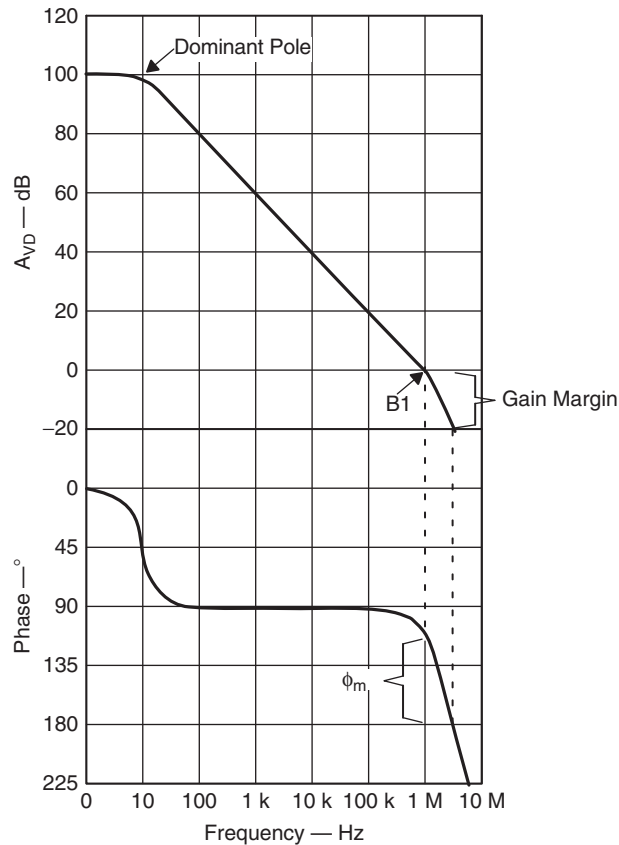


Figure 13.1: Gain and phase margin: Voltage amplification and phase shift versus frequency.

13.6 Open Loop Voltage Gain Parameter, A_{OL}

The open loop voltage gain parameter, A_{OL} , is defined as the ratio of change in output voltage to the change in voltage across the input terminals. Usually, the DC value and a graph showing the frequency dependence are shown in the data sheet. It is expressed either unitless or in decibels.

A_{OL} is similar to the open loop gain, A_{VD} , of the amplifier except that A_{VD} is usually measured with an output load. A_{OL} is usually measured with no load. Both parameters are measured as an open loop. Figure 13.2 indicates that the A_{OL} from 1 kHz to over 40 kHz is over 90 dB for this specific op amp.

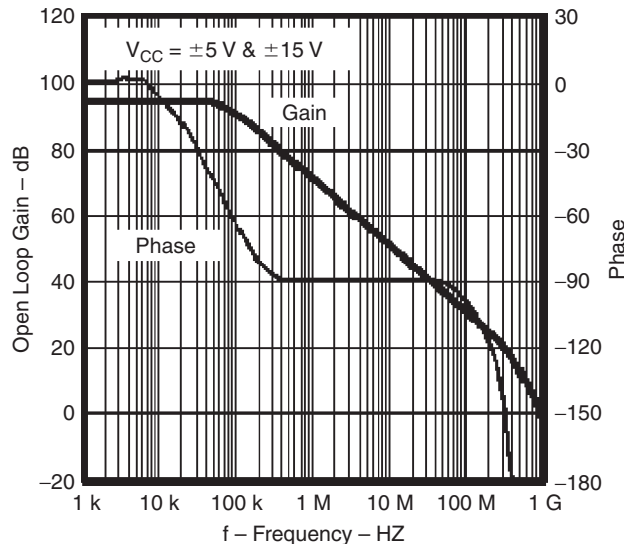


Figure 13.2: Gain and phase margin: Open loop gain and phase response versus frequency.

13.7 Large Signal Voltage Amplification Gain Condition, A_V

The large signal voltage amplification or gain condition, A_V , is defined as the ratio of change in output voltage to the change in voltage across the input terminals that is set up for a test of parameters such as Z_O or THD + N. It is expressed either unitless or in decibels.

13.8 Differential Large Signal Voltage Amplification Parameter, A_{VD}

The differential large signal voltage amplification parameter, A_{VD} , is defined as the ratio of change in output voltage to the change in voltage across the input terminals. It is expressed either unitless or in decibels. A_{VD} is sometimes referred to as *differential voltage gain*.

A_{VD} is similar to the open loop gain, A_{OL} , of the amplifier except that A_{OL} is usually measured with no load and A_{VD} is usually measured with a load. Both parameters are measured as an open loop. Figure 13.1 shows a typical graph of A_{VD} versus frequency.

A_{VD} is a design issue when precise gain is required. This is because the higher the open loop gain for the frequency bandwidth being used, the more precise is the gain. The ratio of resistors in the op amp circuit more closely defines the gain.

13.9 Unity Gain Bandwidth Parameter, B_1

The unity gain bandwidth, B_1 , is defined as the range of frequencies within which the open loop voltage amplification is greater than or equal to unity (0 dB). B_1 is expressed in units of hertz.

13.10 Maximum Output Swing Bandwidth Parameter, B_{OM}

The maximum output swing bandwidth parameter, B_{OM} , is defined as the maximum frequency that the output swing is above a specified value or at the extent of its linear range. B_{OM} is also called *full power bandwidth*. B_{OM} is expressed in units of hertz.

The limiting factor for B_{OM} is the slew rate (SR). As the frequency gets higher and higher, the output becomes slew rate limited and cannot respond quickly enough to maintain the specified output voltage swing.

A simple test that relies on visual acuity for this parameter is to connect a sine wave generator to the input of an op amp circuit with a gain of 1. An oscilloscope or spectrum analyzer can be used to view the output. The generator is adjusted at low frequency and the amplitude increased until distortion (clipping) is visible on the output. The amplitude is then reduced until no distortion is visible. If a specific output amplitude is desired instead of the distortion free amplitude, then the generator should be adjusted to this amplitude. The frequency is increased until distortion is visible, then reduced until again there is no distortion visible. This is the B_{OM} . If the frequency is increased further, the resulting waveform would look more and more like a sawtooth wave. When a sawtooth wave is achieved, increasing the frequency decreases the amplitude.

The following equation expresses the relationship between B_{OM} and SR:

$$B_{OM} = SR/2\pi V_{(PP)}$$

13.11 Bandwidth Parameter, BW

Bandwidth, BW, is defined as the maximum frequency that an op amp circuit can deliver the specified output. The specified output varies and includes conditions such as small signal (-3 dB), 0.1 dB flatness, and full power. BW is expressed in units of hertz.

13.12 Input Capacitance Parameter, C_I

The input capacitance parameter, C_I , is defined as the capacitance between the input terminals of an op amp with either input grounded. It is expressed in units of farads.

C_I is one of a group of parasitic elements affecting input impedance. Figure 13.3 shows a model of the resistance and capacitance between each input terminal and ground and between the two terminals. There is also parasitic inductance, but the effects are negligible at low frequency. Input impedance is a design issue when the source impedance is high. The input loads the source.

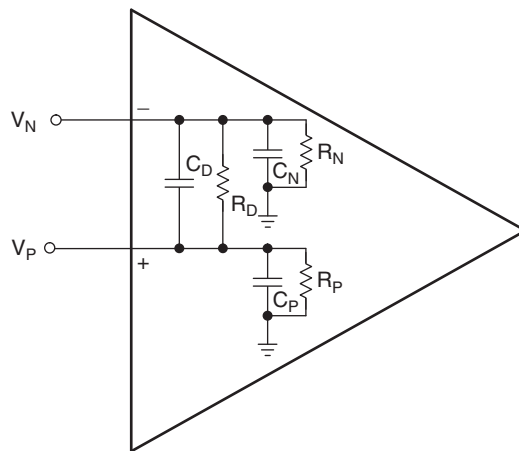


Figure 13.3: Input parasitic elements.

Input capacitance, C_I , is measured between the input terminals with either input grounded. C_I is usually a few picofarads. In [Figure 13.3](#), if V_P is grounded, then $C_I = C_D \parallel C_N$.

Sometimes common mode input capacitance, C_{ic} is specified. In [Figure 13.3](#), if V_P is shorted to V_N , then $C_{ic} = C_P \parallel C_N$. C_{ic} is the input capacitance a common mode source would see when referenced to ground.

13.13 Common Mode Input Capacitance Parameter, C_{ic} or $C_{i(c)}$

The common mode input capacitance parameter, C_{ic} or $C_{i(c)}$, is defined as the input capacitance a common mode source would see to ground. It is expressed in units of farads.

C_{ic} is one of a group of parasitic elements affecting input impedance. [Figure 13.3](#) shows a model of the resistance and capacitance between each input terminal and ground and between the two terminals. There is also parasitic inductance, but the effects are negligible at low frequency. Input impedance is a design issue when the source impedance is high. The input loads the source.

Input capacitance, C_I , is measured between the input terminals with either input grounded. C_I is usually a few picofarads. In [Figure 13.3](#), if V_P is grounded, then $C_I = C_D \parallel C_N$.

Sometimes, common mode input capacitance, C_{ic} is specified. In [Figure 13.3](#), if V_P is shorted to V_N , then $C_{ic} = C_P \parallel C_N$. C_{ic} is the input capacitance a common mode source would see when referenced to ground.

13.14 Differential Input Capacitance Parameter, C_{id}

The differential input capacitance parameter, C_{id} , is the same as the common mode input capacitance, C_{ic} . It is the input capacitance a common mode source would see to ground. It is expressed in units of farads.

13.15 Load Capacitance Condition, C_L

The load capacitance condition, C_L , is defined as the capacitance between the output terminal of an op amp and ground. It is expressed in units of farads.

C_L is a capacitive load that is sometimes connected to an op amp when parameters such as SR, t_s , Φ_m , or A_m are being tested.

13.16 Supply Voltage Sensitivity, $\Delta V_{DD\pm}(\text{or } CC\pm)/\Delta V_{IO}$ or k_{SVS}

The power supply rejection ratio, $\Delta V_{DD\pm}(\text{or } CC\pm)/\Delta V_{IO}$, is the same as the supply rejection ratio, k_{SVR} . It is defined as the absolute value of the ratio of the change in supply voltages to the resulting change in input offset voltage. Typically, both supply voltages are varied symmetrically. It is expressed in decibels.

The power voltage affects the bias point of the input differential pair. Because of the inherent mismatches in the input circuitry, changing the bias point changes the offset voltage, which in turn changes the output voltage.

For a dual supply op amp, $k_{SVR} = \Delta V_{CC\pm}/\Delta V_{OS}$ or $\Delta V_{DD\pm}/\Delta V_{OS}$. The plus and minus sign in the term $\Delta V_{CC\pm}$ means that the plus and minus power supplies are changed symmetrically. For a single supply op amp, $k_{SVR} = \Delta V_{CC}/\Delta V_{OS}$ or $\Delta V_{DD}/\Delta V_{OS}$. Also note that the mechanism that produces k_{SVR} is the same as for the common mode rejection ratio (CMRR). Therefore, k_{SVR} as published in the data sheet is a DC parameter, like CMRR. When k_{SVR} is graphed versus frequency, it falls off as the frequency increases.

Switching power supplies produce noise frequencies from 50 kHz to 500 kHz and higher. The value of k_{SVR} is almost zero at these frequencies, so noise on the power supply results in noise on the output of the op amp. Proper bypassing techniques must be used.

13.17 Common Mode Rejection Ratio Parameter, CMRR or k_{CMR}

The common mode rejection ratio parameter, CMRR or k_{CMR} , is defined as the ratio of differential voltage amplification to common mode voltage amplification. This is measured by determining the ratio of a change in input common mode voltage to the resulting change in input offset voltage. It is expressed in decibels.

Ideally, CMRR or k_{CMR} would be infinite with common mode voltages being totally rejected.

The common mode input voltage affects the bias point of the input differential pair. Because of the inherent mismatches in the input circuitry, changing the bias point changes the offset voltage, which in turn changes the output voltage. The real mechanism at work is $\Delta V_{OS}/\Delta V_{COM}$.

In a Texas Instruments data sheet, $CMRR = \Delta V_{COM}/\Delta V_{OS}$, which gives a positive number in decibels. CMRR, as published in the data sheet, is a DC parameter. CMRR, when graphed versus frequency, falls off as the frequency increases.

A common source of common mode interference voltage is 50 Hz or 60 Hz AC noise. Care must be used to ensure that the CMRR of the op amp is not degraded by other circuit components. High values of resistance make the circuit vulnerable to common mode (and other) noise pick up. It is usually possible to scale resistors down and capacitors up to preserve circuit response.

13.18 Frequency Condition, f

Frequency condition, f , is the frequency available to a circuit for a specific parameter test. It is expressed in hertz.

13.19 Op Amp Gain Bandwidth Product Parameter, GBW

The gain bandwidth product, GBW, is defined as the product of the open loop voltage gain and the frequency at which it is measured. GBW is expressed in units of hertz. [Figure 13.1](#) shows the open loop bandwidth graphically.

The GBW is similar to the unity gain bandwidth (B_1). While B_1 specifies the frequency at which the gain of the op amp is 1, GBW specifies the gain bandwidth product of the op amp at a frequency that may be different than the B_1 .

The GBW is constant for voltage feedback amplifiers. It does not have much meaning for current feedback amplifiers, because there is no linear relationship between gain and bandwidth.

When an op amp is selected for a specific application, both the bandwidth and the slew rate should be taken into account (along with other factors, including power consumption, distortion, and price).

13.20 Supply Current (Shutdown) Parameter, $I_{CC(SHDN)}$ or $I_{DD(SHDN)}$

The supply current (shutdown) parameter, $I_{CC(SHDN)}$ or $I_{DD(SHDN)}$, is defined as the current into the V_{CC+} (V_{DD+}) or V_{CC-} (V_{DD-}) terminal of the amplifier while it is turned off. It is expressed in units of amps.

13.21 Supply Current Parameter, I_{CC} or I_{DD}

The supply current parameter, I_{CC} or I_{DD} , is defined as the current into the V_{CC+} (V_{DD+}) or V_{CC-} (V_{DD-}) terminal of the op amp while it is operating with no load and the input or output is at virtual ground. It is expressed in units of amps.

In a Texas Instruments data sheet, this parameter is usually the total quiescent current draw for the whole package. There are exceptions, however, such as data sheets that cover single and multiple packaged op amps of the same type. In these cases, I_{DD} is the quiescent current draw for each amplifier.

13.22 Input Current Range Parameter, I_I

The input current range parameter, I_I , is defined as the amount of current that can be sourced or sunk by the op amp input. It is usually specified as an absolute maximum rating expressed in units of amperes.

13.23 Input Bias Current Parameter, I_{IB}

The input bias current parameter, I_{IB} , is defined as the average of the currents into the two input terminals with the output at a specified level. It is expressed in units of amperes.

The input circuitry of all op amps requires a certain amount of bias current for proper operation. The input bias current, I_{IB} , is computed as the average of the two inputs:

$$I_{IB} = \frac{I_N + I_P}{2}$$

CMOS and JFET inputs offer much lower input current than standard bipolar inputs. Figure 13.4 shows a typical test circuit for measuring input bias currents.

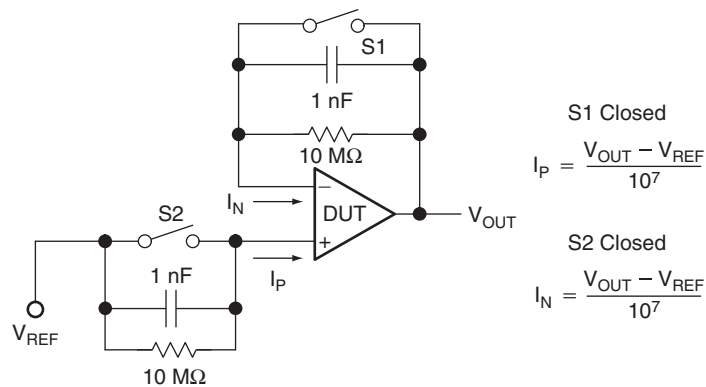


Figure 13.4: Test circuit, I_{IB} .

Input bias current is of concern when the source impedance is high. If the op amp has high input bias current, it loads the source and a lower than expected voltage is seen. If the source impedance is high, the best solution is to use an op amp with either CMOS or JFET input. The source impedance can also be lowered by using a buffer stage to drive the op amp that has high input bias current.

In the case of bipolar inputs, offset current can be nullified by matching the impedance seen at the inputs. In the case of CMOS or JFET inputs, the offset current is usually not an issue and matching the impedance is not necessary.

13.24 Input Offset Current Parameter, I_{IO}

The input offset current parameter, I_{IO} , is defined as the difference between the currents into the two input terminals of an op amp with the output at the specified level. It is expressed in units of amperes.

13.25 Input Noise Current Parameter, I_n

The input noise current parameter, I_n , is defined as the internal noise current reflected back to an ideal current source in parallel with the input pins. It is expressed in units of amperes divided by the square root of hertz.

It is important for a designer to calculate the noise that the device will deliver in an application. The simplest way to calculate this noise is to use the following equation:

$$e_{nt} = \sqrt{V_n^2 + (I_n \times R_s)^2}$$

where

e_{nt} = Total noise voltage.

V_n = Voltage noise (nV/ $\sqrt{\text{Hz}}$).

I_n = Current noise (pA/ $\sqrt{\text{Hz}}$).

R_s = Source resistance (Ω).

13.26 Output Current Parameter, I_O

The output current parameter, I_O , is defined as the amount of current that may be drawn from the op amp output. Usually, I_O is expressed in units of amperes.

Sometimes (for micropower op amps), I_O is specified as an absolute maximum rating, but for most op amps, it is part of the electrical characteristics tables and graphs.

13.27 Low Level Output Current Condition, I_{OL}

The low level output current condition, I_{OL} , is defined as the current into an output that is supplied during the test for V_{OL} . It is usually expressed in units of amperes.

13.28 Short Circuit Output Current Parameter, I_{OS} or I_{SC}

The short circuit output current parameter, I_{OS} or I_{SC} , is defined as the maximum output current available from the amplifier with the output shorted to ground, either supply, or a specified point. Sometimes, a low value series resistor is specified. It is usually expressed in units of amperes.

It is important to observe power dissipation ratings to keep the junction temperature below the absolute maximum rating when the output is heavily loaded or shorted. See the absolute maximum ratings section of the part's data sheet for more information.

13.29 Supply Rejection Ratio Parameter, k_{SVR}

The supply rejection ratio, k_{SVR} , is the same as the power supply rejection ratio, PSRR. It is defined as the absolute value of the ratio of the change in supply voltages to the resulting change in input offset voltage. Typically both supply voltages are varied symmetrically. It is expressed in decibels.

The power voltage affects the bias point of the input differential pair. Because of the inherent mismatches in the input circuitry, changing the bias point changes the offset voltage, which in turn changes the output voltage.

For a dual supply op amp, $k_{SVR} = \Delta V_{CC\pm}/\Delta V_{OS}$ or $\Delta V_{DD\pm}/\Delta V_{OS}$. The plus and minus sign in the term $\Delta V_{CC\pm}$ means that the plus and minus power supplies are changed symmetrically. For a single supply op amp, $k_{SVR} = \Delta V_{CC}/\Delta V_{OS}$ or $\Delta V_{DD}/\Delta V_{OS}$. Also note that the mechanism that produces k_{SVR} is the same as for CMRR. Therefore, k_{SVR} as published in the data sheet is a DC parameter, like CMRR. When k_{SVR} is graphed versus frequency, it falls off as the frequency increases.

Switching power supplies produce noise frequencies from 50 kHz to 500 kHz and higher. The value of k_{SVR} is almost zero at these frequencies, so noise on the power supply results in noise on the output of the op amp. Proper bypassing techniques must be used.

13.30 Power Dissipation Parameter, P_D

The power dissipation, P_D , is defined as the power supplied to the device less any power delivered from the device to a load. Note: At no load, $P_D = V_{CC+} \times I_{CC}$ or $P_D = V_{DD+} \times I_{DD}$. It is expressed in units of watts.

13.31 Power Supply Rejection Ratio Parameter, PSRR

The power supply rejection ratio, PSRR, is the same as the supply rejection ratio, k_{SVR} . It is defined as the absolute value of the ratio of the change in supply voltages to the resulting change in input offset voltage. Typically, both supply voltages are varied symmetrically. It is expressed in units of decibels.

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Switching power supplies produce noise frequencies from 50 kHz to 500 kHz and higher. The value of k_{SVR} is almost zero at these frequencies, so noise on the power supply results in noise on the output of the op amp. Proper bypassing techniques must be used.

13.32 Junction to Ambient Thermal Resistance Parameter, θ_{JA}

The junction to the ambient thermal resistance parameter, θ_{JA} , is defined as the ratio of the difference in temperature from the die junction to the ambient air and the power dissipated by the die. The value of θ_{JA} is expressed in units of degrees Centigrade per watt.

The term θ_{JA} is dependent on the case to the ambient thermal resistance as well as the θ_{JC} parameter.

The θ_{JA} is a better indicator of thermal resistance when the package is not well thermally sunk to other components in the assembly.

The θ_{JA} is listed in the data sheet for different packages. It is useful for evaluating which package is least likely to overheat and determine what the die temperature is when the ambient temperature and power dissipation are known.

The following equation more fully defines the math of heat dissipation:

$$T_J = T_A + PD \times \theta_{JA}$$

where

$$\theta_{JA} = \theta_{JC} + \theta_{CH} + \theta_{HA}$$

T_A (C°) = Temperature of ambient air.

T_J (C°) = Temperature of semiconductor junction.

- PD (W) = Power dissipated in semiconductor.
- θ_{JC} ($^{\circ}\text{C}/\text{W}$) = Thermal resistance (junction to case).
- θ_{CH} ($^{\circ}\text{C}/\text{W}$) = Thermal resistance (case to heat sink).
- θ_{HA} ($^{\circ}\text{C}/\text{W}$) = Thermal resistance (heat sink to ambient air).
- θ_{JA} ($^{\circ}\text{C}/\text{W}$) = Thermal resistance (junction to ambient air).

The design of a heat sink is based on empirical measurements made by the heat sink manufacturer and made within the enclosure of the final system. The actual math is straightforward. It looks like a simple electrical circuit. Take temperature to be the equivalent of voltage; thermal resistance is now just a simple resistor and the power is considered the same as current. Now, this looks like a linear circuit, and in fact, the resistances add linearly.

Figure 13.5 compares the performance of two heat sinks, each at two power levels. A place to start is at ground or the ambient temperature. Since the temperature within an enclosure or in some environmental areas can range widely, the expected maximum temperature should be used for T_A .

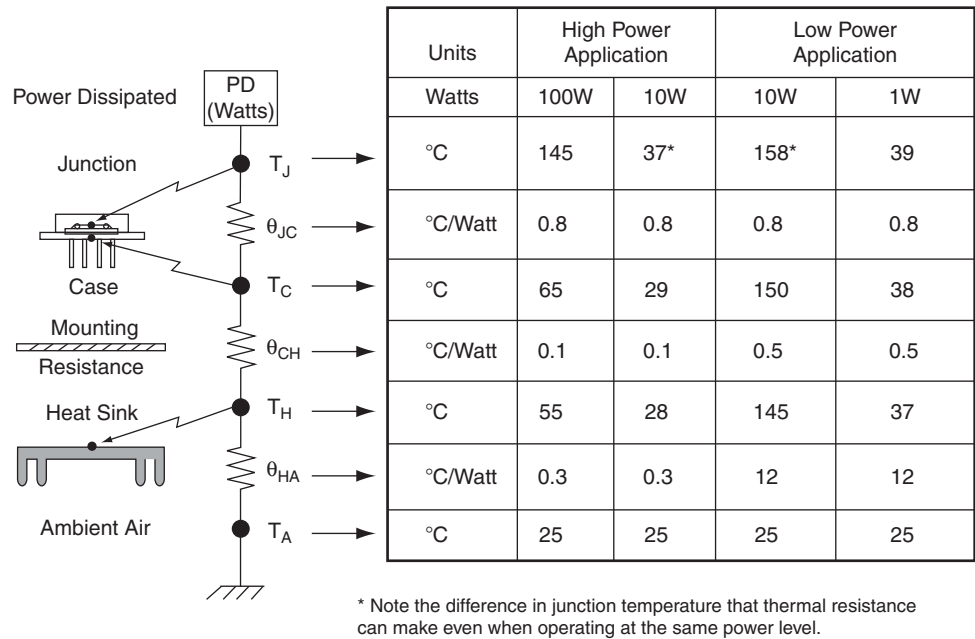


Figure 13.5: Thermal resistances.

The power is multiplied by the heat sink thermal resistance and added to the ambient temperature to find the temperature at the interface between the heat sink and the case. Each step is the same as this one, finding the temperature at the next point.

Now, the power has to pass through the mounting resistance of either 0.1 or 0.5°C/W. The 10 W dissipation can result in a junction temperature of either 37°C or 158°C. This points out the advantage of using a highly efficient heat sink to get the heat out of the device and into the air.

Installing a fan on the heat sink significantly increases its effectiveness. For example, nearly all PCs use a fan attached to the processor's heat sink for this reason.

13.33 Junction to Case Thermal Resistance Parameter, θ_{JC}

The junction to the case thermal resistance parameter, θ_{JC} , is defined as the ratio of the difference in temperature from the die junction to the case and the power dissipated by the die. The value of θ_{JC} is expressed in units of degrees Centigrade per watt.

The term θ_{JC} is not dependent on the case to the ambient thermal resistance, as is the θ_{JA} parameter. The θ_{JC} is a better indicator of thermal resistance when the package is to be thermally sunked to other components in the assembly.

The θ_{JC} is listed in the data sheet for different packages. It is useful for evaluating which package is least likely to overheat and determine what the die temperature is when the case temperature and power dissipation are known.

The following equation more fully defines the math of heat dissipation:

$$T_J = T_A + PD \times \theta_{JA}$$

where

$$\theta_{JA} = \theta_{JC} + \theta_{CH} + \theta_{HA}$$

T_A (C°) = Temperature of ambient air.

T_J (C°) = Temperature of semiconductor junction.

PD (W) = Power dissipated in semiconductor.

θ_{JC} ($^{\circ}\text{C}/\text{W}$) = Thermal resistance (junction to case).

θ_{CH} ($^{\circ}\text{C}/\text{W}$) = Thermal resistance (case to heat sink).

θ_{HA} ($^{\circ}\text{C}/\text{W}$) = Thermal resistance (heat sink to ambient air).

θ_{JA} ($^{\circ}\text{C}/\text{W}$) = Thermal resistance (junction to ambient air).

The design of a heat sink is based on empirical measurements made by the heat sink manufacturer and made within the enclosure of the final system. The actual math is straightforward. It looks like a simple electrical circuit. Take temperature to be the equivalent of voltage; thermal resistance is now just a simple resistor and the power is considered the same as current. Now, this looks like a linear circuit, and in fact, the resistances add linearly.

Figure 13.5 compares the performance of two heat sinks, each at two power levels. A place to start is at ground or at the ambient temperature. Since the temperature within an enclosure or in some environmental areas can range widely, the expected maximum temperature should be used for T_A .

The power is multiplied by the heat sink thermal resistance and added to the ambient temperature to find the temperature at the interface between the heat sink and the case. Each step is the same as this one, finding the temperature at the next point.

Now, the power has to pass through the mounting resistance of either 0.1 or $0.5^{\circ}\text{C}/\text{W}$. The 10 W dissipation can result in a junction temperature of either 37°C or 158°C . This points out the advantage of using a highly efficient heat sink to get the heat out of the device and into the air.

Installing a fan on the heat sink significantly increases its effectiveness. For example, nearly all PCs use a fan attached to the processor's heat sink for this reason.

13.34 Input Resistance Parameter, r_i

The input resistance parameter, r_i , is defined as the DC resistance between the input terminals with either input grounded. It is expressed in units of ohms.

The r_i is one of a group of parasitic elements affecting input impedance. Figure 13.3 shows a model of the resistance and capacitance between each input terminal and

ground and between the two terminals. There is also parasitic inductance, but the effects are negligible at low frequency. Input impedance is a design issue when the source impedance is high. The input loads the source.

Input resistance, r_i , is the resistance between the input terminals with either input grounded. In Figure 13.3, if V_P is grounded, then $r_i = R_D \parallel R_N$. The value of r_i ranges from $10^7 \Omega$ to $10^{12} \Omega$, depending on the type of input.

Sometimes common mode input resistance, r_{ic} , is specified. In Figure 13.3, if V_P is shorted to V_N , then $r_{ic} = R_P \parallel R_N$. The r_{ic} is the input resistance a common mode source would see when referenced to ground.

13.35 Differential Input Resistance Parameter (r_{id} or $r_{i(d)}$)

The differential input resistance, r_{id} or $r_{i(d)}$, is defined as the small signal resistance between two ungrounded input terminals. It is expressed in units of ohms.

The r_{id} is one of a group of parasitic elements affecting input impedance. Figure 13.3 shows a model of the resistance and capacitance between each input terminal and ground and between the two terminals. There is also parasitic inductance, but the effects are negligible at low frequency. Input impedance is a design issue when the source impedance is high. The input loads the source.

Input resistance, r_i , is the resistance between the input terminals with either input grounded. In Figure 13.3, if V_P is grounded, then $r_i = R_D \parallel R_N$. The value of r_i ranges from $10^7 \Omega$ to $10^{12} \Omega$, depending on the type of input.

Sometimes, common mode input resistance, r_{ic} , is specified. In Figure 13.3, if V_P is shorted to V_N , then $r_{ic} = R_P \parallel R_N$. The r_{ic} is the input resistance a common mode source would see when referenced to ground. In Figure 13.3, $r_{id} = R_D$.

13.36 Load Resistance Condition, R_L

The load resistance condition, R_L , is defined as the DC resistance that is attached from the output of an op amp to ground during a test for a parameter such as A_{VD} , SR, THD + D, $t_{(on)}$, $t_{(off)}$, GBW, t_s , ϕ_m , or A_m . It is expressed in units of ohms.

13.37 Null Resistance Condition, R_{null}

The null resistance condition, R_{null} , is defined as the DC resistance attached in series with C_L when testing for parameters such as phase margin or gain margin. It is expressed in units of ohms.

13.38 Output Resistance Parameters, r_o

The output resistance parameter, r_o , is defined as the DC resistance placed in series with the output of an ideal amplifier and the output terminal for simulation of the real device. It is expressed in units of ohms.

This parameter is difficult to measure, because when an amplifier is configured as an open loop, it is hard to balance the inputs to cause the output to become zero (or $V_{CC}/2$ if single supply) because of the extremely high open loop gain. When this is accomplished, a load current can be drawn from the output and the change in voltage measured. Ohm's law is then used to calculate r_o .

13.39 Signal Source Condition, R_S

The signal source condition, R_S , is defined as the output resistance of a signal source. It is expressed in units of ohms.

R_S is used as a test condition when measuring parameters such as V_{IO} , α_{VIO} , I_{IO} , I_{IB} , or CMMR. A typical value for R_S used in these parameter tests is 50 Ω .

13.40 Open Loop Transresistance Parameters, R_t

In a transimpedance or current feedback amplifier, the open loop transresistance parameter, R_t , is defined as the ratio of change in DC output voltage to the change in DC current at the inverting input. It is expressed in units of ohms.

13.41 Op Amp Slew Rate Parameter, SR

The slew rate parameter, SR, is defined as the rate of change in the output voltage caused by a step change at the input. It is expressed in volts per second. The SR parameter of an op amp is the maximum SR it will pass and is generally specified with a gain of 1. [Figure 13.6](#) shows SR graphically.

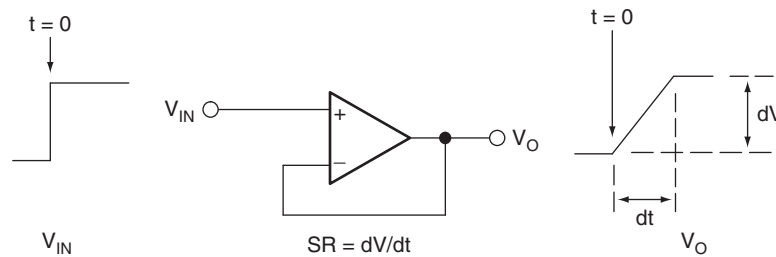


Figure 13.6: Slew rate.

For an amplifier to pass a signal without distortion due to insufficient SR, the amplifier must have at least the maximum SR of the signal. The maximum SR of a sine wave occurs as it crosses zero. The following equation defines this slew rate:

$$SR = 2\pi f V$$

f = Frequency of the signal.

V = Peak voltage of the signal.

The SR is sometimes represented as $SR+$ and $SR-$. $SR+$ is the abbreviation for the slew rate for a positive transition, and $SR-$ is the abbreviation for the slew rate for a negative transition. Many applications are best served when $SR+$ and $SR-$ are the same magnitude.

The primary factor controlling SR in most op amps is an internal compensation capacitor, which is added to make the op amp unity gain stable. Remember, however, that not all op amps have compensation capacitors. In op amps without internal compensation capacitors, the SR is determined by internal op amp parasitic capacitances. Noncompensated op amps generally exhibit faster SR than decompensated op amps, and decompensated op amps exhibit faster slew rates than fully compensated op amps. When noncompensated or decompensated op amps are used, the designer must ensure the stability of the circuit by other means.

The type of op amp that offers the fastest SR is the current feedback op amp. It is optimized for fast SR. Figure 13.7 shows a table of the bandwidth and SR of two high speed op amps. The graph shows the step response of a current feedback (CFB) versus a voltage feedback (VFB) op amp.

CHARACTERISTIC Part number	VOLTAGE FEEDBACK	CURRENT FEEDBACK
	OPA2652	THS3001
Bandwidth (MHz)	700	420
Slew rate (V/ μ s)	335	6500

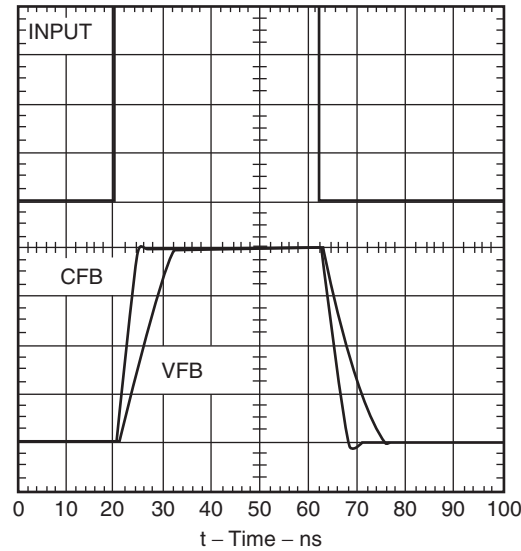


Figure 13.7: Bandwidth versus slew rate.

When an op amp is selected for a specific application both the bandwidth and the SR should be taken into account (along with other factors, including power consumption, distortion, and price).

13.42 Operating Free Air Temperature Condition, T_A

The operating free air temperature condition, T_A , is defined as the free air temperature over which the op amp is being operated. Some of the other parameters may change with temperature, leading to degraded operation at temperature extremes. The value of T_A is expressed in units of degrees Centigrade.

A range of T_A is listed in a data sheet's absolute maximum ratings table, because stress beyond those listed may cause permanent damage to the device. Functional operation

to this limit is not implied and may affect reliability. Another range of T_A is listed in a data sheet's recommended operating conditions. The term T_A is also used in a data sheet to define a condition during parameter tests and typical characteristic graphs. In addition, it may be one of the axis variables on a chart.

13.43 Turn off Time (Shutdown) Parameter, t_{DIS} or $t_{(off)}$

The turn off time (shutdown) parameter, t_{DIS} or $t_{(off)}$, is defined as the time from when the turn off voltage is applied to the shutdown pin to when the supply current has reached half of its final value. It is expressed in units of seconds.

13.44 Turn on Time (Shutdown) Parameter, t_{EN}

The turn on time (shutdown) parameter, t_{EN} , is defined as the time from when the turn on voltage is applied to the shutdown pin to when the supply current has reached half of its final value. It is expressed in units of seconds.

13.45 Fall Time Parameter, t_f

The fall time parameter, t_f , is defined as the time required for an output voltage step to change from 90% to 10% of its final value. It is expressed in units of seconds.

13.46 Total Harmonic Distortion Parameter, THD

The total harmonic distortion parameter, THD, is defined as the ratio of the rms voltage of the harmonics of the fundamental signal to the total rms voltage at the output. THD is expressed in “dBc” (decibel relative to the carrier) or percent.

THD does not account for the noise as does the total harmonic distortion plus noise parameter.

13.47 Total Harmonic Distortion Plus Noise Parameter, $THD + N$

The total harmonic distortion plus noise parameter, $THD + N$, is defined as the ratio of the rms noise voltage plus the rms harmonic voltage of the fundamental signal to the fundamental rms voltage signal at the output. It is expressed in dBc or percent.

THD + N compares the frequency content of the output signal to the frequency content of the input. Ideally, if the input signal is a pure sine wave, the output signal is a pure sine wave. Due to nonlinearity and noise sources within the op amp, the output is never pure.

To simplify further, THD + N is the ratio of all other frequency components to the fundamental:

$$\text{THD} + \text{N} = \left[\frac{(\sum \text{Harmonic voltages} + \text{Noise voltages})}{\text{Fundamental}} \right] \times 100\%$$

Figure 13.8 shows a hypothetical graph where $\text{THD} + \text{N} = 1\%$. The fundamental is the same frequency as the input signal. The nonlinear behavior of the op amp results in harmonics of the fundamental being produced in the output. The noise in the output is due to the input noise of the op amp and other noise sources. All the harmonics and noise added together are 1% of the fundamental.

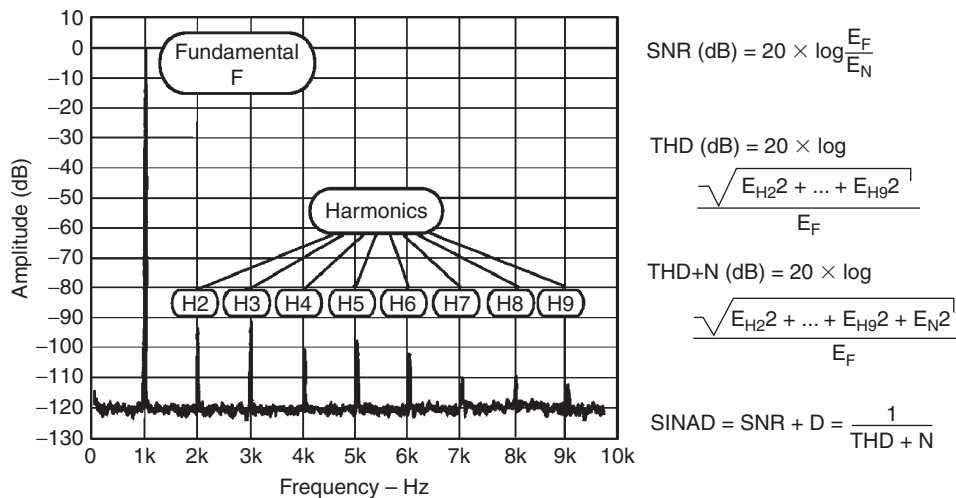


Figure 13.8: Graph of THD + N = 1%.

The three greatest causes of distortion in an op amp are the limit on output voltage swing, Class A–B crossover nonlinearity, and slew rate. Typically an op amp must be

operated at or below its recommended operating conditions to realize low THD. Figure 13.9 shows two signals. The lower frequency signal is a low distortion sine wave while the higher frequency signal is distorted because of slew rate limitation. Figure 13.10 shows a signal with clipping caused by a limited output voltage swing. Finally, Figure 13.11 shows a signal with crossover distortion caused by Class A–B crossover nonlinearity.

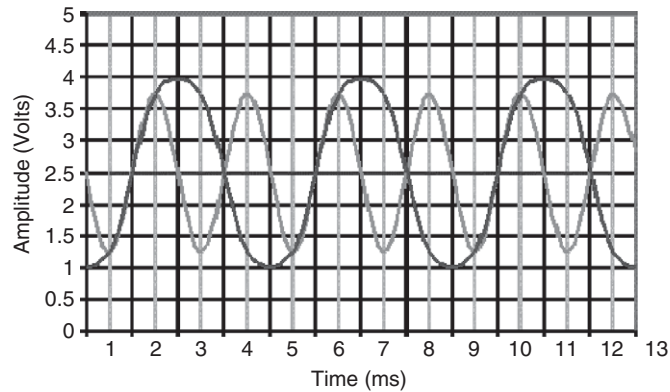


Figure 13.9: Effect of slew rate limits on signals.

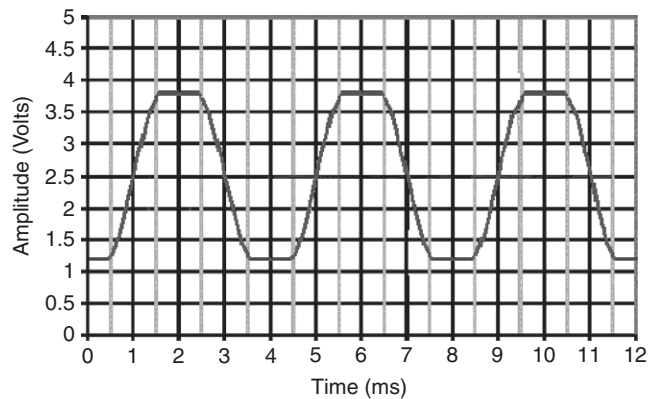


Figure 13.10: Signal with distortion due to clipping.

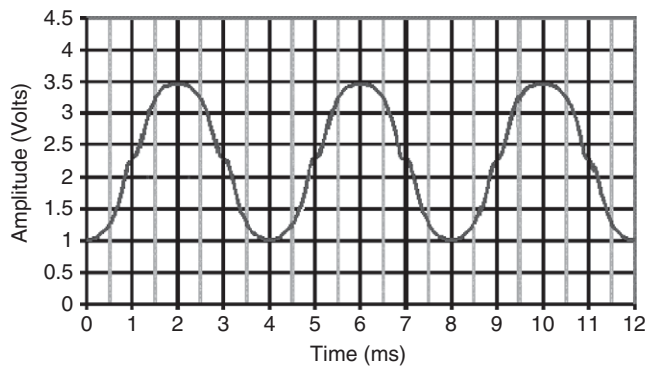


Figure 13.11: Signal with crossover distortion.

13.48 Maximum Junction Temperature Parameter, T_J

The maximum junction temperature parameter, T_J , is defined as the temperature over which the die may be operated. Some of the other parameters may change with temperature, leading to degraded operation at temperature extremes. The value of T_J is expressed in units of degrees Centigrade.

The term T_J is listed in the absolute maximum ratings table, because stress beyond those listed may cause permanent damage to the device. Functional operation to this limit is not implied and may affect reliability.

13.49 Rise Time Parameter, t_r

The rise time parameter, t_r , is defined as the time required for an output voltage step to change from 10% to 90% of its final value. It is expressed in units of seconds.

13.50 Settling Time Parameter, t_s

The settling time parameter, t_s , is defined as the time required for the output voltage to settle within the specified error band of the final value with a step change at the input. It is also known as *total response time*, t_{tot} . It is expressed in units of seconds.

It takes a finite time for a signal to propagate through the internal circuitry of an op amp. Therefore, it takes a period of time for the output to react to a step change in the input. In addition, the output normally overshoots the target value, experiences damped oscillation, and settles to a final value. Figure 13.12 shows t_s graphically.

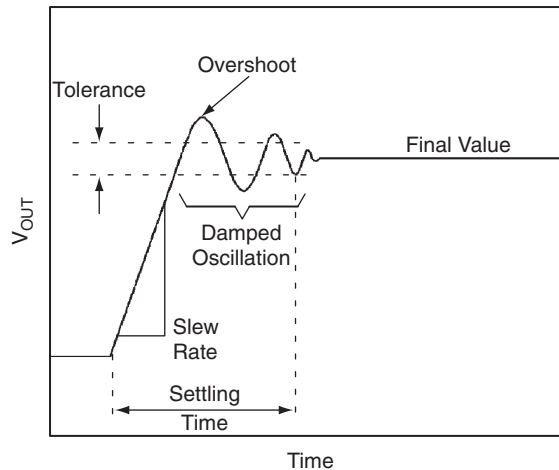


Figure 13.12: Settling time.

Settling time is a design issue in data acquisition circuits when signals are changing rapidly. An example is when using an op amp following a multiplexer to buffer the input to an A to D converter. Step changes can occur at the input to the op amp when the multiplexer changes channels. The output of the op amp must settle to within a certain tolerance before the A to D converter samples the signal.

13.51 Storage Temperature Parameter, T_S or T_{stg}

The storage temperature parameter, T_S or T_{stg} , is defined as the temperature over which the op amp may be stored (unpowered) for long periods of time without damage. It is expressed in units of degrees Centigrade.

13.52 Supply Voltage Condition, V_{CC} or V_{DD}

The supply voltage condition, V_{CC} or V_{DD} , is defined as the bias voltage applied to the op amp power supply pin(s). For single supply applications, it is specified as a positive value; and for split supply applications, it is specified as a plus or minus value, referenced to analog ground. It is expressed in units of volts.

The term V_{CC} or V_{DD} is often defined in the maximum ratings, recommended operating conditions, and as a test condition in parameter tables and graphs, because the voltage

supplied has an important impact on the way a circuit operates. It is also used as one of the axis variables in some of the characteristic graphs.

13.53 Input Voltage Range Condition or Parameter, V_I

The input voltage range parameter, V_I , is defined as the range of input voltages that may be applied to either the $IN+$ or $IN-$ inputs.

The input voltage range condition, V_I , is defined as the voltage delivered to a circuit input when testing for V_O on a graph, such as Large Signal Inverting Pulse Response versus Time.

The value of V_I is expressed in units of volts for either a condition or parameter.

13.54 Common Mode Input Voltage Condition, V_{IC}

The common mode input voltage condition, V_{IC} , is defined as the voltage that is common to both input pins. The value of V_{IC} as expressed in units of volts.

The term V_{IO} set at $V_{DD}/2$ (for single supply op amps) is often used as a condition when testing for various parameters including V_{IO} , I_{IO} , I_{IB} , V_{OH} , or V_{OL} .

When a two wire signal is subject to noise and this noise is being received equally on both signal lines, it can be rejected by a differential amplifier with good common mode rejection.

13.55 Common Mode Input Voltage Range Parameter, V_{ICR}

The common mode input voltage range parameter, V_{ICR} , is defined as the range of common mode input voltage that, if exceeded, may cause the operational amplifier to cease functioning properly. This sometimes is taken as the voltage range over which the input offset voltage remains within a set limit. The value of V_{ICR} is expressed in units of volts.

The input common voltage, V_{IC} , is defined as the average voltage at the inverting and noninverting input pins. If the common mode voltage gets too high or too low, the inputs shut down and proper operation ceases. The common mode input voltage range, V_{ICR} , specifies the range over which normal operation is guaranteed.

Different input structures allow for different input common mode voltage ranges:

- The LM324 and LM358 use bipolar pnp inputs that have their collectors connected to the negative power rail. This allows the common mode input voltage range to include the negative power rail.
- The TL07X and TLE207X type BiFET op amps use p-channel JFET inputs with the sources tied to the positive power rail via a bipolar current source. This allows the common mode input voltage range to include the positive power rail.
- TI LinCMOS op amps use p channel CMOS inputs with the substrate tied to the positive power rail. This allows the common mode input voltage range to include the negative power rail.
- Rail to rail input op amps use complementary n- and p-type devices in the differential inputs. When the common mode input voltage nears either rail, at least one of the differential inputs is still active, and the common mode input voltage range includes both power rails.

The trends toward lower, single supply voltages make V_{ICR} of increasing concern.

Rail to rail input is required when a noninverting unity gain amplifier is used and the input signal ranges between both power rails. An example of this is the input of an analog to digital converter in a low voltage, single supply system.

High side sensing circuits require operation at the positive input rail.

13.56 Differential Input Voltage Parameter, V_{ID}

The differential input voltage parameter, V_{ID} , is defined as the voltage at the noninverting input with respect to the inverting input. The value of V_{ID} is expressed in units of volts.

The limit V_{ID} is usually defined in the absolute maximum ratings table, because stress beyond this limit may cause permanent damage to the device.

13.57 Differential Input Voltage Range Parameter, V_{DIR}

The input common mode voltage range parameter, V_{DIR} , is defined as the range of differential input voltage that, if exceeded, may cause the operational amplifier to cease functioning properly. The value of V_{DIR} is expressed in units of volts.

Some devices have protection built into them, and the current into the input needs to be limited. Normally, differential input mode voltage limit is not a design issue.

13.58 Turn on Voltage (Shutdown) Parameter, $V_{IH-SHDN}$ or $V_{(ON)}$

The turn on voltage (shutdown) parameter, $V_{IH-SHDN}$ or $V_{(ON)}$, is defined as the voltage required on the shutdown pin to turn the device on. It is expressed in units of volts.

13.59 Turn off Voltage (Shutdown) Parameter, $V_{IL-SHDN}$ or $V_{(OFF)}$

The turn off voltage (shutdown) parameter, $V_{IL-SHDN}$ or $V_{(OFF)}$, is defined as the voltage required on the shutdown pin to turn the device off. It is expressed in units of volts.

13.60 Input Voltage Condition, V_{IN}

The input voltage condition, V_{IN} , is defined as the DC voltage delivered to a circuit input when testing for V_n . The value of V_{IN} is expressed in units of volts.

13.61 Input Offset Voltage Parameter, V_{IO} or V_{OS}

The input offset voltage parameter, V_{IO} or V_{OS} , is defined as the DC voltage that must be applied between the input terminals to cancel DC offsets within the op amp. It is expressed in units of volts.

All op amps require a small voltage between their inverting and noninverting inputs to balance mismatches due to unavoidable process variations. The required voltage, known as the *input offset voltage*, is abbreviated V_{IO} . V_{IO} is normally modeled as a voltage source driving the noninverting input.

Figure 13.13 shows two typical methods for measuring input offset voltage. DUT stands for device under test. Test circuit (a) is simple, but since V_{OUT} is not at zero volts, it does not really meet the definition of the parameter. Test circuit (b) is referred to as a servo loop. The action of the loop is to maintain the output of the DUT at zero volts.

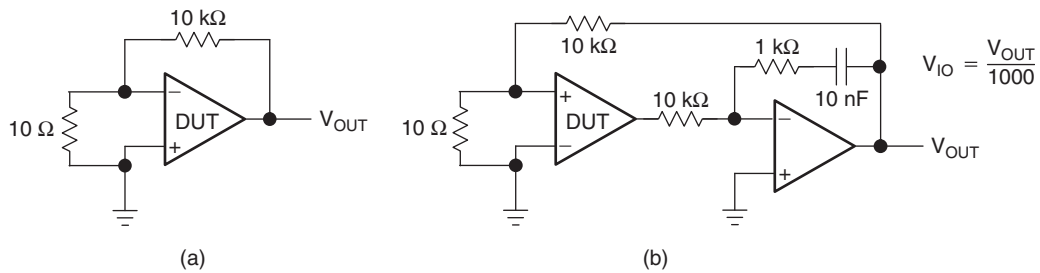


Figure 13.13: Test circuits for input offset voltage.

V_{IO} is an input referred parameter. This means that it is amplified by the positive closed loop gain of the circuit. For this reason, the figure indicates that V_{IO} is V_{OUT} divided by 1000. This constant is used because the gain of the DUT is 1000.

V_{IO} is normally attributed to the characteristics of the input differential pair in a voltage feedback amplifier. Different processes provide certain advantages. Bipolar input stages tend to have lower offset voltages than CMOS or JFET input stages.

Input offset voltage is of concern anytime that DC accuracy is required of the circuit. One way to null the offset is to use external null inputs on a single op amp package (see [Figure 13.14](#)). A potentiometer is connected between the null inputs with the adjustable terminal connected to the negative supply through a series resistor. The input offset voltage is nulled by shorting the inputs and adjusting the potentiometer until the output is zero.

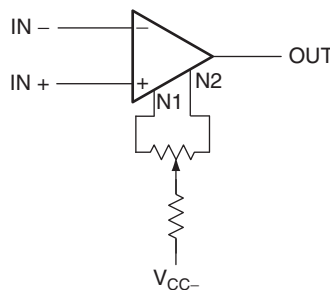


Figure 13.14: Offset voltage adjust.

13.62 Equivalent Input Noise Voltage Parameter, V_n

The equivalent input noise voltage parameter, V_n , is defined as the internal noise voltage reflected back to an ideal voltage source in parallel with the input pins at a specific frequency. The value of V_n is expressed in units of volts per the square root of hertz.

When this parameter is measured, the noise measured at the output (with the input connected to virtual ground) is divided by the gain of the amplifier circuit. This is the amplitude of noise at the input that would be amplified by an ideal amplifier to cause an equivalent signal at the output.

The limit of V_n is sometimes defined at several frequencies in the operating characteristics table or as a graph.

Given the same structure within an op amp, increasing bias currents lowers noise (and increases SR, GBW, and power dissipation).

Also the resistance seen at the input to an op amp adds noise. Balancing the input resistance on the noninverting input to that seen at the inverting input, while helping with offsets due to input bias current, adds noise to the circuit.

It is important for a designer to calculate noise that the device will deliver in an application. The simplest way is to calculate this noise is to use the following equation:

$$e_{nt} = \sqrt{V_n^2 + (I_n \times R_s)^2}$$

where

e_{nt} = Total noise voltage.

V_n = Voltage noise (nV/ $\sqrt{\text{Hz}}$).

I_n = Current noise (pA/ $\sqrt{\text{Hz}}$).

R_s = Source resistance (Ω).

13.63 Broadband Noise Parameter ($V_{N(\text{PP})}$)

The broadband noise parameter, $V_{N(\text{PP})}$, is defined as the peak to peak voltage over a specific frequency band, typically 0.1 Hz to 1 Hz or 0.1 Hz to 10 Hz. The units of measurement are typically volts peak to peak.

Given the same structure within an op amp, increasing bias currents lowers noise (and increases SR, GBW, and power dissipation).

Also the resistance seen at the input to an op amp adds noise. Balancing the input resistance on the noninverting input to that seen at the inverting input, while helping with offsets due to input bias current, adds noise to the circuit.

13.64 High Level Output Voltage Condition or Parameter, V_{OH}

The high level output voltage parameter, V_{OH} , is defined as the positive rail of the op amp output for the load current conditions applied to the power pins. When the V_{OH} parameter is tested, it may be defined with an I_{OH} of -1 mA, -20 mA, -35 mA, or -50 mA load.

When V_{OH} is listed on a data sheet as a test condition, it is used for testing another parameter. Whether V_{OH} is a condition or a parameter it is expressed in units of volts.

13.65 Low Level Output Voltage Condition or Parameter, V_{OL}

The low level output voltage parameter, V_{OL} , is defined as the negative rail of the op amp output for the load current conditions applied to the power pins. When the V_{OL} parameter is tested it may be defined with an I_{OL} of -1 mA, -20 mA, -35 mA, or -50 mA load.

When V_{OL} is listed on a data sheet as a test condition, it is used for testing another parameter.

Whether V_{OL} is a condition or parameter it is expressed in units of volts.

13.66 Maximum Peak to Peak Output Voltage Swing Parameter, $V_{OM\pm}$

The maximum peak to peak output voltage swing parameter, $V_{OM\pm}$, is defined as the maximum peak to peak output voltage that can be obtained without clipping when the op amp is operated from a bipolar supply. The value of $V_{OM\pm}$ is expressed in units of volts.

Another way to describe $V_{OM\pm}$ is the maximum positive or negative peak output voltage that can be obtained without waveform clipping, when quiescent DC output voltage is zero. $V_{OM\pm}$ is limited by the output impedance of the amplifier, the saturation voltage of the output transistors, and the power supply voltages. This is shown pictorially in Figure 13.15.

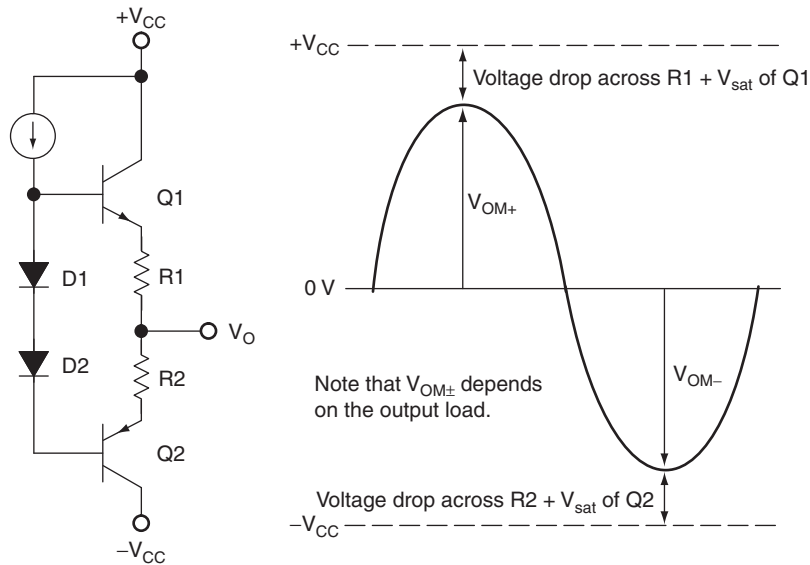


Figure 13.15: $V_{OM\pm}$.

This emitter/follower structure cannot drive the output voltage to either rail. Rail to rail output op amps use a common emitter (bipolar) or common source (CMOS) output stage. With these structures, the output voltage swing is limited by only the saturation voltage (bipolar) or the on resistance (CMOS) of the output transistors and the load being driven.

Because newer products are focused on single supply operation, more recent data sheets from Texas Instruments use the terminology V_{OH} and V_{OL} to specify the maximum and minimum output voltages.

Maximum and minimum output voltage is usually a design issue when dynamic range is lost if the op amp cannot drive to the rails. This is the case in single supply systems where the op amp is used to drive the input of an A to D converter, which is configured for full scale input voltage between ground and the positive rail.

13.67 Peak to Peak Output Voltage Swing Condition or Parameter, $V_{O(PP)}$

The peak to peak output voltage swing condition, $V_{O(PP)}$, is defined as the peak to peak voltage set up on the output waveform to test for parameters such as A_{VD} or SR.

The peak to peak output voltage swing parameter, $V_{O(PP)}$, is the maximum peak to peak output voltage that an op amp can deliver. When it is measured, V_{DD} , THD + N, R_L , and T_A are the typical test conditions.

The value of $V_{O(PP)}$ is also expressed in units of volts for either a condition or parameter.

13.68 Step Voltage Peak to Peak Condition, $V_{(STEP)PP}$

The step voltage peak to peak condition, $V_{(STEP)PP}$, is defined as the peak to peak voltage step that is used as a test condition for parameters such as t_s . The value of $V_{(STEP)PP}$ is expressed in units of volts.

13.69 Crosstalk Parameter, X_T

The crosstalk parameter, X_T , is defined as the ratio of the change in output voltage of a driven channel to the resulting change in output voltage from another channel that is not driven. The value of X_T is expressed in units of decibels.

X_T is a function of how good the separation is between channels in an IC package or system. It is caused by the signal from one channel being coupled to the other channel inductively, capacitively, through the power supply, and so forth.

13.70 Output Impedance Parameter, Z_o

The output impedance parameter, Z_o , is defined as the frequency dependent small signal impedance that is placed in series with an ideal amplifier and the output terminal in a closed loop configuration. The value of Z_o is expressed in units of ohms.

Figure 13.16 is an example of defining Z_o and is shown with gains of 1, 10, and 100.

Common emitter (bipolar) and common source (CMOS) output stages used in rail to rail output op amps have higher output impedance than emitter/follower output stages.

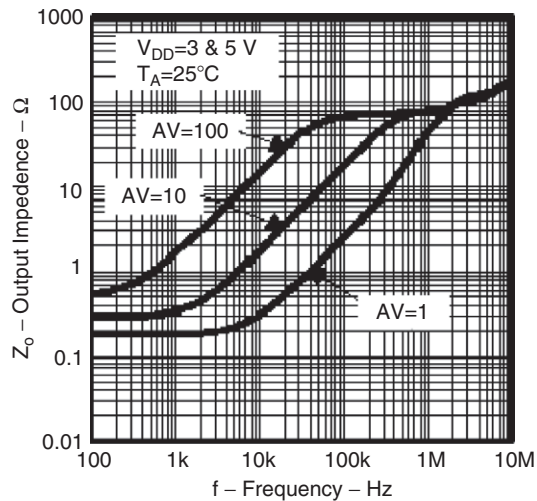


Figure 13.16: Output impedance versus frequency.

Output impedance is a design issue when using rail to rail output op amps to drive heavy loads. If the load is mainly resistive, the output impedance limits how close to the rails the output can go. If the load is capacitive, the extra phase shift erodes the phase margin.

Figure 13.17 shows how output impedance affects the output signal, assuming Z_o is mostly resistive.

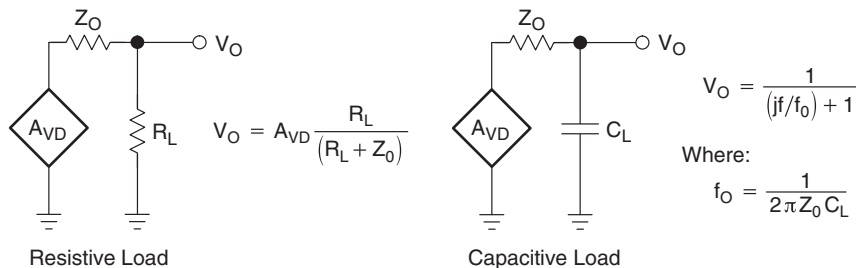


Figure 13.17: Effect of output impedance.

Some new audio op amps are designed to drive the load of a speaker or headphone directly. They can be an economical method of obtaining very low output impedance.

13.71 Open Loop Transimpedance Parameter, Z_t

The open loop transimpedance parameter, Z_t , is defined as the frequency dependent ratio of change in output voltage to the frequency dependent change in current at the inverting input in a transimpedance or current feedback amplifier. The value of Z_t is expressed in units of ohms.

13.72 Differential Phase Error Parameter, Φ_D

The differential phase error parameter, Φ_D , is defined as the change in AC phase with a change in DC level. The AC signal is 40 IRE (0.28 V PK) and the DC level change is ± 100 IRE (± 0.7 V). It is typically tested at 3.58 MHz (NTSC) or 4.43 MHz (PAL) carrier frequencies. The value of Φ_D is expressed in units of degrees.

13.73 Phase Margin Parameter, Φ_m

The phase margin parameter, Φ_m , is defined as the absolute value of the difference in the phase shift of 180° and the phase shift at unity gain. The Φ_m is measured as an open loop and is expressed in units of degrees:

$$\Phi_m = 180^\circ - \Phi \text{ at } B_1$$

Gain margin (A_m) and phase margin (Φ_m) are different ways of specifying the stability of the circuit. Since rail to rail output op amps have higher output impedance, a significant phase shift is seen when driving capacitive loads. This extra phase shift erodes the phase margin, and for this reason, most CMOS op amps with rail to rail outputs have limited ability to drive capacitive loads.

Figure 13.1 shows Φ_m graphically.

13.74 Bandwidth for 0.1 dB Flatness

The bandwidth for 0.1 dB flatness is defined as the range of frequencies within which the gain is ± 0.1 dB of the nominal value with full output power. It is expressed in units of hertz. Figure 13.18 describes a device with the bandwidth for 0.1 dB flatness to 100 MHz.

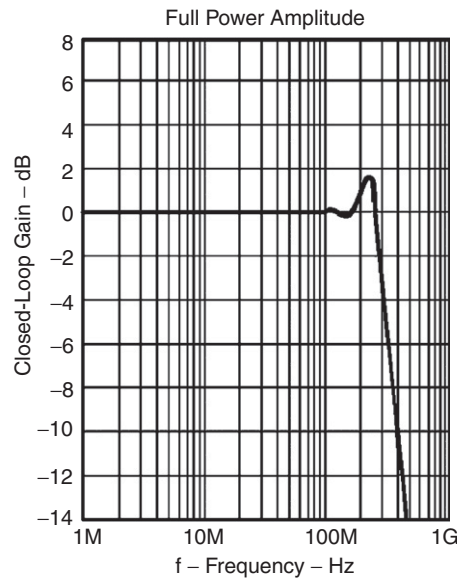


Figure 13.18: 0.1 dB flatness.

13.75 Case Temperature for 60 Seconds

The case temperature for 60 s is defined as the temperature to which the case may safely be exposed for 60 s. It is usually specified as an absolute maximum and is meant as a guide for automated soldering processes. It is expressed in degrees Centigrade.

13.76 Continuous Total Dissipation Parameter

The continuous total dissipation parameter is defined as the power that can be dissipated by an op amp package, including loads. It is usually specified as an absolute maximum. This parameter may be broken down by ambient temperature and package style in a table.

Continuous total dissipation is expressed in units of watts.

13.77 Duration of Short Circuit Current

The duration of the short circuit current parameter is defined as the amount of time that the output can be shorted to network ground. It is usually specified as an absolute maximum. The duration of short circuit current is usually expressed in seconds.

13.78 Input Offset Voltage Long Term Drift Parameter

The input offset voltage long term drift parameter is defined as the ratio of the change in input offset voltage to the change in time. It is the average value for the month and is expressed in units of volts per month.

13.79 Lead Temperature for 10 or 60 Seconds

The lead temperature for 10 or 60 s is defined as the temperature to which the leads may safely be exposed for 10 or 60 s. It is usually specified as an absolute maximum and is meant as a guide for automated soldering processes. This parameter is expressed in units of degrees Centigrade.